Nested Congruence Constraints on Collatz Synchronization

Paper H: Jump Heights, Odd-Core Transport, and the Consecutive Cascade Conflict

Hiroki Kamanoi

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Abstract

Building on the odd-core decomposition $n_t = 1 + 2^{a_t} q_t$ (Paper G), we establish the algebraic mechanism governing synchronization jump heights. At a renewal event ($a_t = 1$), the next jump height is determined by the residue of q_t:

$$a_{t+1} = v_2(3q_t + 1), \quad \text{and} \quad a_{t+1} \geq K \iff q_t \equiv r_K \pmod{2^K},$$

where $r_K = -3^{-1} \bmod 2^K$ is the unique odd residue satisfying $3r_K \equiv -1 \pmod{2^K}$.

We prove **Theorem JE** (Jump-Explosion Link): the jump formula is exact, and achievability of height K requires q_t to lie in an exponentially thin residue class of density 2^{-K}.

We then prove **Theorem CC** (Consecutive Cascade Conflict): two consecutive jumps of the same height $K \geq 4$ are impossible. The proof uses the known order $\mathrm{ord}_{2^K}(3) = 2^{K-2}$ and the fact that the cascade length $L = \lfloor (K-2)/2 \rfloor$ satisfies $0 < L < 2^{K-2}$, so $3^L \not\equiv 1 \pmod{2^K}$.

We identify the core open problem for Paper I: can jumps of *increasing* heights $K_1 < K_2 < K_3 < \cdots \to \infty$ be simultaneously achievable by a single orbit, given that each cascade multiplies q by $3^{\lfloor K_i/2 \rfloor}$?

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1 Introduction

Paper G established that each synchronization cascade of height k multiplies the odd core q by $3^{\lfloor (k-2)/2 \rfloor}$. The present paper asks: given this forced growth of q, what jump heights can subsequently be achieved?

The answer requires translating the jump condition $a_{t+1} \geq K$ into a congruence condition on q_t. The key formula is:

$$a_{t+1} = v_2(3q_t + 1) \quad \text{when } a_t = 1.$$

This reduces the question of achievable jump heights to a residue problem: what residue classes modulo 2^K does q_t visit?

1.1 Structure of the paper

Section 2 proves Theorem JE (the jump formula and residue condition). Section 3 proves Theorem CC (consecutive cascade conflict). Section 4 analyzes the nested congruence chain for sequences of jumps. Section 5 formulates the core open problem.

2 Theorem JE: Jump Formula

Theorem 2.1 (JE — Jump-Explosion Link)**.** *At a renewal event $a_t = 1$ (i.e., $n_t \equiv 3 \pmod 4$), write $n_t = 1 + 2q_t$ with q_t odd. Then:*

1. Jump formula*: $a_{t+1} = v_2(3q_t + 1)$.*
2. Height-K condition*: $a_{t+1} \geq K$ if and only if $q_t \equiv r_K \pmod{2^K}$, where*

$$r_K := -3^{-1} \bmod 2^K \in \{1, 3, 5, \ldots, 2^K - 1\}$$

   is the unique odd residue satisfying $3r_K \equiv -1 \pmod{2^K}$.
3. Density*: the residue class $\{q : q \equiv r_K \pmod{2^K}\}$ has density 2^{-K} in the odd integers.*

Proof. **Part (1).** With $n_t = 1 + 2q_t$:

$$3n_t + 1 = 4 + 6q_t = 2(2 + 3q_t).$$

Since q_t is odd, $2 + 3q_t$ is odd, so $v_2(3n_t + 1) = 1$. Thus $T(n_t) = 2 + 3q_t$ and

$$T(n_t) - 1 = 1 + 3q_t, \quad a_{t+1} = v_2(1 + 3q_t) = v_2(3q_t + 1). \quad \checkmark$$

Part (2). $a_{t+1} \geq K \iff v_2(3q_t+1) \geq K \iff 2^K \mid 3q_t+1 \iff 3q_t \equiv -1 \pmod{2^K}$. Since $\gcd(3, 2^K) = 1$, this is equivalent to $q_t \equiv -3^{-1} \equiv r_K \pmod{2^K}$.

Part (3). The residue class $r_K \bmod 2^K$ among odd integers has density 2^{-K}: one out of every 2^{K-1} odd residues.

Oddness of r_K. Since $3 \cdot r_K \equiv -1 \pmod 2$ and 3 is odd, we need $r_K \equiv 1 \pmod 2$, i.e., r_K is odd. □ □

Remark 2.2 (Connection to Theorem JH)**.** Theorem JH (Paper F) stated that under Haar measure, $P(a_{t+1} = k) = 2^{-k}$. This is now explained mechanically: achieving jump height exactly k means $q_t \equiv r_k \pmod{2^k}$ but $q_t \not\equiv r_{k+1} \pmod{2^{k+1}}$, which has probability $2^{-k} - 2^{-(k+1)} = 2^{-(k+1)}$... [see numerical observation for exact values].

The values of r_K follow the pattern:

K	r_K	$r_K \bmod 2^K$ (condition on q_t)
1	1	$q_t \equiv 1 \pmod 2$ (always, q_t odd)
2	1	$q_t \equiv 1 \pmod 4$
3	5	$q_t \equiv 5 \pmod 8$
4	5	$q_t \equiv 5 \pmod{16}$
5	21	$q_t \equiv 21 \pmod{32}$
6	21	$q_t \equiv 21 \pmod{64}$
7	85	$q_t \equiv 85 \pmod{128}$
K	r_K	density 2^{-K}

3 Theorem CC: Consecutive Cascade Conflict

Lemma 3.1 (Order of 3 mod 2^K)**.** *For $K \geq 3$, $\mathrm{ord}_{2^K}(3) = 2^{K-2}$.*

Proof. This is a standard result in 2-adic number theory. The group $(\mathbb{Z}/2^K\mathbb{Z})^\times$ has order 2^{K-1} for $K \geq 3$, and 3 generates its cyclic subgroup of index 2, giving order 2^{K-2}. □

Theorem 3.2 (CC — Consecutive Cascade Conflict)**.** *Let $K \geq 4$. Suppose a Collatz orbit achieves a jump of height K at renewal time t (i.e., $a_t = 1$, $a_{t+1} = K$). Let $L = \lfloor (K-2)/2 \rfloor$ be the cascade length. Then a* second *jump of height $\geq K$ at the next renewal time $t + L$ (assuming $a_{t+L} = 1$) is* ***impossible****.*

Proof. **Setup.** By Theorem JE, the first jump requires:

$$q_t \equiv r_K \pmod{2^K}. \tag{C1}$$

By Corollary (Cascade formula, Paper G), after L steps of cascade:

$$q_{t+L} = 3^L q_t. \tag{C2}$$

For a second height-K jump, Theorem JE requires:

$$q_{t+L} \equiv r_K \pmod{2^K}. \tag{C3}$$

Combining (C1)–(C3):

$$3^L r_K \equiv r_K \pmod{2^K} \implies (3^L - 1) r_K \equiv 0 \pmod{2^K}.$$

Since r_K is odd (Theorem JE, part 2), $\gcd(r_K, 2^K) = 1$, so:

$$3^L \equiv 1 \pmod{2^K}. \tag{C4}$$

Contradiction. By Lemma 3.1, $\mathrm{ord}_{2^K}(3) = 2^{K-2}$. Condition (C4) requires $2^{K-2} \mid L$. But for $K \geq 4$:

$$L = \lfloor \frac{K-2}{2} \rfloor \leq \frac{K-2}{2} < K - 2 \leq 2^{K-2},$$

where the last inequality holds since $K - 2 < 2^{K-2}$ for all $K \geq 4$ (easily verified). Thus $0 < L < \mathrm{ord}_{2^K}(3)$, which prevents $3^L \equiv 1 \pmod{2^K}$. □ □

Remark 3.3 (Numerical verification)**.** The impossibility of same-height consecutive jumps for $K = 4, \ldots, 9$ was verified by direct computation over all odd residues modulo 2^{15}, confirming zero cases.

Corollary 3.4. *Between any two jumps of height $\geq K$ (for $K \geq 4$), the jump sequence must include at least one jump of height $< K$. In other words, the synchronization level cannot stay at or above K across consecutive renewal cycles.*

4 The Nested Congruence Chain

Theorem CC shows that the same height cannot repeat immediately. We now analyze the conditions for a sequence of increasing jump heights $K_1 < K_2 < K_3 < \cdots$.

Proposition 4.1 (Nested congruence chain)**.** *Suppose a Collatz orbit achieves jump heights $K_1, K_2, \ldots, K_n$ in sequence (at renewal times $\tau_1 < \tau_2 < \cdots < \tau_n$). Then the initial odd core q_{τ_1} must satisfy the nested congruence system:*

$$q_{\tau_1} \equiv r_{K_1} \pmod{2^{K_1}}, \tag{1}$$

$$3^{L_1} q_{\tau_1} \equiv r_{K_2} \pmod{2^{K_2}}, \tag{2}$$

$$3^{L_1 + L_2} q_{\tau_1} \equiv r_{K_3} \pmod{2^{K_3}}, \tag{3}$$

$$\vdots$$

where $L_i = \lfloor (K_i - 2)/2 \rfloor$ is the cascade length of the i-th jump.

Proof. Equation (1) is Theorem JE. Equation (2): by Paper G's cascade formula, $q_{\tau_2} = q_{\tau_1 + L_1} = 3^{L_1} q_{\tau_1}$. Theorem JE applied to the second jump gives $3^{L_1} q_{\tau_1} \equiv r_{K_2}$. The pattern continues by induction. □

Proposition 4.2 (CRT analysis)**.** *The system (1)–(3) is solvable by CRT if and only if for each pair (i, j) with $K_i = K_j$:*

$$r_{K_i} \cdot 3^{-(L_1 + \cdots + L_{i-1})} \equiv r_{K_j} \cdot 3^{-(L_1 + \cdots + L_{j-1})} \pmod{\gcd(2^{K_i}, 2^{K_j})}.$$

By Theorem CC, all same-height pairs are incompatible. For strictly increasing heights $K_1 < K_2 < \cdots$, the congruences are modulo distinct powers 2^{K_i}, and CRT guarantees compatibility—but the combined residue class has density $\prod_i 2^{-K_i}$, which tends to 0 as $n \to \infty$.

5 The Core Open Problem

Open Problem 5.1 (Infinite nested congruence achievability)**.** Does there exist an odd integer q_0 such that the Collatz orbit starting at $n_0 = 1 + 2q_0$ achieves jump heights $K_1 < K_2 < K_3 < \cdots \to \infty$?

Equivalently: does there exist q_0 satisfying the infinite nested congruence system

$$3^{S_i} q_0 \equiv r_{K_i} \pmod{2^{K_i}}, \quad S_i = \sum_{j<i} L_j,$$

for some strictly increasing sequence $K_1 < K_2 < \cdots$?

Remark 5.2 (Why this matters)**.** By Paper E (Theorem SE), $\sup_K R_K < \infty$ is equivalent to Collatz convergence. By Paper F (Theorem JH and the renewal model), $\sup_t a_t = \infty$ (2-adic recurrence) requires arbitrarily high jump heights. Open Problem 5.1 asks whether any single orbit can achieve this.

The congruence density argument shows: the n-th condition has density 2^{-K_n}. The combined density of the first n conditions is $\prod_{i=1}^{n} 2^{-K_i} \to 0$. This does not prove impossibility (a specific orbit is deterministic, not random), but it quantifies the "arithmetic cost" of achieving all conditions simultaneously.

6 Summary

Result	Status	Section
Theorem JE: $a_{t+1} = v_2(3q_t + 1)$	**Proved**	§2
Theorem JE: height-K iff $q_t \equiv r_K \pmod{2^K}$	**Proved**	§2
Lemma: $\mathrm{ord}_{2^K}(3) = 2^{K-2}$	Known	§3
Theorem CC: same-height consecutive jumps impossible ($K \geq 4$)	**Proved**	§3
Nested congruence chain	**Proved**	§4
CRT compatibility analysis	Proved	§4
Open: infinite nested congruences achievable?	**Open**	§5

References

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